

Coral Reef Bleaching: A 2024 Field Summary (Test Document)

This is a fabricated test document for the Claude API course, used to exercise PDF support and citations. The numbers below are illustrative, not real research data.

Background

Coral bleaching occurs when sustained ocean temperature increases cause corals to expel the symbiotic algae living in their tissues, draining the coral of its color and a major energy source. Repeated or prolonged bleaching events raise coral mortality significantly.

Survey Findings

Across the twelve reef sites surveyed in this exercise, average sea surface temperature was 1.8 degrees Celsius above the ten-year seasonal baseline. Seven of the twelve sites recorded bleaching coverage above 40 percent of live coral cover. The two sites with the highest water flow showed the lowest bleaching coverage, at 12 percent and 15 percent respectively, suggesting that circulation may offer some thermal buffering.

Recovery Observations

Sites revisited after ninety days showed partial pigment recovery in roughly a third of previously bleached colonies, while the remaining colonies either died or remained bleached. Recovery correlated most strongly with depth: colonies below eight meters recovered at roughly twice the rate of shallow colonies.

Recommendation

The summary recommends prioritizing monitoring at shallow, low-flow sites, since this exercise's data shows they combine the highest bleaching incidence with the lowest recovery rate.